DERIVED LIMITS OF BOUNDED-BELOW INVERSE SYSTEMS EXIST WITH COUNTABLE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10733-10739, immediately following the statement at 10726-10731.

Editorial note. The sufficiency reading of the Rlim definition is the span's own step.

Allowed standard prerequisites (editorial).

- Injective resolutions exist. Lemma lemma-injective-resolutions-exist of this chapter, cited at proof lines 10735-10736. Each K_n is represented by a bounded below complex I_n of injectives.
- Products of K-injective complexes. Lemma lemma-product-K-injective of this chapter, cited at proof lines 10737-10739. The product of K-injective complexes represents the product in D(A) and is K-injective.

Source reference. Original lemma 013K: sources/stacks-derived/pinned-original-source.tex, source lines 6497–6497. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BK6: sources/stacks-derived/pinned-original-source.tex, source lines 9917–9917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with countable products and enough injectives. Let (K_n) be an inverse system of $D^+(\mathcal{A})$. Then $R\lim K_n$ exists.*

Proof. It suffices to show that $\prod K_n$ exists in $D(\mathcal{A})$. For every n we can represent K_n by a bounded below complex $I_n^\bullet$ of injectives (Lemma 013K). Then $\prod K_n$ is represented by $\prod I_n^\bullet$, see Lemma 0BK6. □

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